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Summary

Data & ML Platform Engineer with 3+ years designing production-grade distributed data systems processing 10+ TB/day. Track record at Uber (120+ datasets, 99.9% uptime, 30% cloud cost optimization) and Synchron (100+ enterprise clients). Deep expertise in Apache Spark, Kafka, Airflow, AWS, and Python. Hands-on experience in MLOps, NLP, LLM fine-tuning, RAG pipelines, agentic AI workflows, and model serving using PyTorch, LangChain, and Hugging Face. Targeting Data Engineer / ML Engineer / GenAI Engineer roles.

Technical Skills

Languages: Python, SQL, Java, Shell Scripting

Big Data & ETL: Apache Spark, Kafka, Flink, Hive, Presto/Trino, Airflow (MWAA), dbt, Stream Processing, Data Modeling

Cloud & DevOps: AWS (S3, EMR, Glue, Redshift, Kinesis, SageMaker, Bedrock, Lambda), GCP (BigQuery), Docker, Kubernetes, Terraform, CI/CD Pipelines

ML & MLOps: PyTorch, TensorFlow, scikit-learn, XGBoost, MLflow, Feature Engineering, Model Serving, Model Monitoring, Experiment Tracking, Hyperparameter Tuning

GenAI & NLP: LangChain, Hugging Face Transformers, RAG Pipelines, LLM Fine-tuning (QLoRA/DPO), Prompt Engineering, Agentic AI, FAISS, Pinecone, Embeddings, Semantic Search

Databases: PostgreSQL, MySQL, DynamoDB, Redshift, BigQuery, Delta Lake, Apache Iceberg, Data Warehousing

Analytics & Tools: Tableau, Power BI, Pandas, NumPy, Statistical Analysis, A/B Testing, Git, JIRA, Linux, Databricks

Professional Experience

Software Engineer — Data Platform

Synchron, Bengaluru

May 2025 – Present

Pricing Data Lake Team

- **Architected cloud-native pricing data lake** for 100+ enterprise clients, enabling downstream analytics, ML-driven pricing optimization, and GenAI-powered business insights
- **Cut dashboard latency 87%** (40 min → 5 min) by materializing complex views into optimized physical tables with incremental refresh and columnar storage
- Orchestrated end-to-end Airflow (MWAA) DAGs and AWS Glue PySpark ETL for multi-tenant pricing data ingestion and transformation
- Implemented DMS Serverless migration; built observability framework with data lineage tracking, data quality validation, and anomaly alerting

Software Development Engineer I (Uber)

Nineleaps (Client: Uber Technologies), Bengaluru

Jun 2024 – May 2025

Data Platform

- **Skip-level promotion** (MTS II → SDE I); won “Feather on the Hat” 6x for delivery excellence and production ownership
- **Managed 120+ production datasets** including Tier-1 tables; achieved zero downtime with 99.9% pipeline uptime SLA across distributed systems
- **Reduced cloud infrastructure costs 30%** via Spark optimization, partition pruning, query performance tuning, and EMR cluster right-sizing
- Led incident management on-call for TB-scale pipelines; built monitoring for data freshness, schema drift, and observability

Data Engineer (Uber)

Nineleaps (Client: Uber Technologies), Bengaluru

Sep 2023 – Jun 2024

Data Engineering

- **Built 50+ production ETL pipelines** (Spark, Glue, Airflow) processing 10+ TB/day with stream and batch processing
- **Improved query performance 45%** by redesigning data warehouse schemas, implementing columnar storage, and partition optimization
- Automated data quality monitoring reducing incidents 60%; designed feature store pipelines for fraud detection production ML models

- Built ingestion pipelines (Python, SQL) processing millions of records; created Tableau/Power BI dashboards for cross-functional stakeholders; optimized SQL queries for 50% reporting speedup

Projects

GenAI Document Q&A System with Agentic RAG Pipeline | Python, LangChain, FAISS, Hugging Face, FastAPI, Docker, AWS EC2

- **Architected production-grade RAG application** using LangChain orchestration + FAISS vector database + Mistral-7B LLM for multi-document PDF question-answering with source attribution and context grounding
- Built end-to-end NLP pipeline: document chunking, sentence-transformer embedding generation, semantic similarity search, and hallucination guardrails with evaluation metrics
- Applied prompt engineering (chain-of-thought, few-shot) reducing hallucinations 35%; deployed as Docker microservice with FastAPI REST API and CI/CD via GitHub Actions

Real-Time Fraud Detection ML Pipeline with MLOps | Kafka, Spark Streaming, XGBoost, MLflow, Grafana, Prometheus, Docker

- **Designed end-to-end stream processing system:** Kafka ingestion → Spark Structured Streaming → XGBoost classifier (F1: 0.94) processing 50K+ events/min in real-time
- Engineered 25+ ML features from transaction velocity, geolocation, and merchant behavior; MLflow for experiment tracking, model versioning, and automated retraining on data drift
- Built production model monitoring dashboard (Grafana + Prometheus) tracking model serving latency, prediction drift, throughput, and SLA compliance

Scalable Data Lakehouse with Data Governance on AWS | Spark, Delta Lake, Airflow, dbt, Redshift, Terraform, Great Expectations

- **Designed medallion architecture** (Bronze/Silver/Gold) on S3 + Delta Lake supporting batch and incremental processing from 5+ heterogeneous data sources
- Implemented dbt transformation layer with automated testing, data lineage tracking, and schema evolution; Great Expectations for data quality validation and profiling
- Orchestrated with Airflow DAGs; Terraform IaC provisioning; Redshift BI integration; achieved 99.5% pipeline reliability with quality gates at each layer

Indic Language Chatbot — LLM Fine-Tuning & Model Serving | PyTorch, Hugging Face, QLoRA, DPO, vLLM, Weights & Biases, Gradio

- **Fine-tuned Mistral-7B** with QLoRA (4-bit quantization) on Hindi conversational data; applied DPO preference optimization for RLHF-style human alignment
- Deployed model serving with vLLM (3x inference speedup); W&B experiment tracking for 50+ runs with hyperparameter tuning and model evaluation
- 82% task completion on benchmarks; interactive multi-turn Gradio demo for Indic-language customer support automation

Certifications

Databricks Certified Data Engineer Associate • DeepLearning.AI – LangChain for LLM Application Development

AWS Certified ML – Specialty (In Progress) • **Google Cloud Professional ML Engineer** (In Progress)

Education

Bachelor of Technology in Computer Science

GITAM Deemed University, India

2019 – 2023

Key Achievements

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- **99.9% pipeline uptime** across 120+ datasets at Uber (zero downtime) • 87% latency reduction at Syncron (40 min → 5 min)
 - **30% cloud cost optimization** via Spark tuning and EMR right-sizing • 6x “Feather on the Hat” awards at Uber
 - **Skip-level promotion** (MTS II → SDE I) within 9 months • 10+ TB/day processed across 100+ enterprise clients